Flow sa Website - this need more details!

Name:

-Welcome
-Assessment
-Turn based Game (Still in earth doing basic astro things) - timtim
-Graduation
-Roles and Tradeoffs
-Moon or Mars
-Spaceship going to that Planet
-First Scenario (oxygen levels down) - Applying the knowledge of the training
-Planet Landing
- MIssions in 30 days
- 2D, turnbased, among us type of tasks
- After 30 days, last assessment before going back to earth (if dili kapasar naay penalty bale 3-5 days extension)
-If napasar makauli ka sa earth, maghatagreport then evaluation, then recommendation on the next mission run
-ending

Specifications
Characters and Roles:
Spaceship Repairman
Farmer
Traveller/Scavenger
Researcher/Evaluator
Assessment Part:
Which areas do we need to focus on being an astronaut?
-Machineries/Equipment
-Basic Astro Competency (moon walk etc)
-Farming
-Travelling
-Researcher
-Repair
-Basic reporting and communication

Missions on each roles at least 3 to 5 each day per role in the 30 days trial:
Day 1:
1
2
3

Day 2:

EXPEDITION 33: BASE COMMAND
2026 NASA Space Apps Challenge - "Junior Astro Trainer App"
Game Mechanics & Timeline Flow Plan

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1. PROJECT GOAL & MAIN IDEA
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Problem:
Space educational games make astronaut engineering compromises too basic without drawbacks or way too complex like datasheets to keep students interested.

Solution:
Turn-based 2D web app game where players complete Earth training before running a Moon or Mars station. Styled in 2D environments with turn-based mechanics, custom action cards, quick-time reaction prompts, and dialogue setup.

NASA Data Setup:
Resource values, output levels, planet variables, and hazard stats are handled via dynamic settings ready to connect directly to official NASA data sources.

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2. THE FOUR SYSTEM CONFLICTS
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Across the training run and planetary survival stages, every choice creates interconnected trade-offs across four critical base systems:

1. POWER SUPPLY
   - Role: Generated via solar grids to keep habitat modules operational.
   - Drawback: Cleaning or fixing solar panels takes valuable crew hours and life support oxygen.

2. LIFE SUPPORT (Air & Water Systems)
   - Role: Controls clean air and water recycling inside the base environment.
   - Drawback: Filtering air and purifying water drains heavy power from the main system.

3. FARMING (Hydroponic Greenhouse)
  - Role: Yields food supplies for the team while helping scrub cabin air.
  - Drawback: Running grow lights and water pumps drains both power and water reserves.

4. RADIATION SHIELDING & STRUCTURAL HULL
  - Role: Protects the team against space radiation, solar storms, and surface storms.
  - Drawback: Refilling liquid shield barriers and patching leaks burns power and water stores.

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3. COMPLETE GAME SEQUENCE FLOW
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PHASE 1: Comic Introduction Storyboard

- Story Opening: Starts with a comic book layout instead of standard menus.
- Click Progress: Each click loads the next styled panel with audio effects and dialogues:
  - Panel 1: Becoming an astronaut heading to Moon or Mars bases.
  - Panel 2: Showing why missions crash when power, life support, farming, and shielding clash.
  - Panel 3: Introducing the Earth Training Facility where astronauts handle trade-offs before launch.
  - Panel 4: Putting on the helmet to start training tasks.
- Screen Switch: Finishing the panel scene switches over to the Main Menu.

PHASE 2: Main Menu Setup

Set against a 2D deep space backdrop with menu option panels:

1. START GAME (Locked Initially)
  - Clicking this first alerts players that Earth training is required before mission launch.
  - Unlocks completely once tutorial tasks are done, displaying user stats and perks.

2. TRAINING (Active Option)
  - Begins the Earth simulator to learn turn-based game balance.

3. OPTIONS
  - Adjust sound volume, modify QTE timing limits, or wipe current save files.

4. ABOUT & CREDITS
  - Summary of project design, team roster, and NASA data source list.

PHASE 3: Mandatory Earth Simulator

- 2D Facility Layout:
  Loads an Earth analog dome environment mimicking space terrain with playable astronaut character sprite.

- Turn Mechanics & Tasks:
  - Action Order Bar: Displays action turns along with upcoming simulation hazards.
  - Tactical Action Cards: Hovering displays previews of resource increases and drops before selection.
  - Quick-Time Reaction Prompts: Timing actions correctly grants bonus efficiency while lowering resource penalties.
  - Guided Tutorials: Flight leads guide players through initial scenarios covering Power, Oxygen, Water, and Shielding.

PHASE 4: Results & Evaluation Screen

- Award Scene Transition:
  Clearing training leads directly to an evaluation ceremony screen.
- Grade Score:
  Ranks player execution (Rank S, A, B) based on stat management and QTE accuracy.
- Profile Save Data:
  Stores performance perks for the main run and unlocks the main menu START option.

PHASE 5: Moon or Mars Main Expedition

Selecting the unlocked START button begins the space mission:

- Role and Location Setup:
  - Pick a Specialist Class (Technician, Farmer, Shield Expert, Researcher) with specific traits.
  - Pick Location (Moon Base vs Mars Colony) with distinct environmental weather factors.
  - Applies saved bonus boosts from training directly to Day 1 gauges.

- Base Map & Daily Turn Cycle:
  - 2D Base Layout: Map shows 5 key rooms (Solar Array, Life Support, Hydroponics, Shield Dome, Science Lab) with animated astronaut movement.
  - Daily Loop: Each day provides clear targets and action points to split across base stations.
  - Action Options: Selecting a room displays safe low-risk actions or risky boost choices.
  - Environment Events: Ending a turn consumes core daily supplies and triggers hazards like solar storms or dust clouds.
  - Flight Log Updates: Short text reports explain why status bars shifted after every cycle.

PHASE 6: Mission Summary & Certificate

- Final Performance Check:
  Evaluates if all critical system meters remained safe throughout the target timeline duration.
- Ending Report & Badges:
  Displays mission outcome status, stats summary, unlocked achievement badges, and restart button for alternate runs.